

Adaptive Ensemble Learning for Short-Panel Policy Evaluation: Evidence from Soda Taxes

Lukas Nitsche¹ and Elio Zhang¹

¹Seattle University

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Abstract

Evaluating a local policy often requires choosing among several plausible comparison-group estimators using only a few years of pre-treatment data—too few for cross-validation to choose among them reliably. We propose perturbed cross-validation (PCV), which adds mean-preserving perturbations to the outcomes, repeats the pre-treatment validation comparison many times, and weights the candidate estimators by how often each one performs best. PCV behaves like ordinary cross-validation when a single estimator is clearly best and combines estimators when the evidence is noisy, and we give conditions under which combining lowers prediction error. Applying PCV to San Francisco’s 2018 sugar-sweetened beverage tax, we find no detectable effect on adult obesity: the estimated post-tax gap is small and statistically indistinguishable from gaps in untreated geographic units, in line with prior estimates for the same policy.

Keywords: Policy evaluation; short panels; cross-validation; ensemble methods; soda taxes.

1 Introduction

Many applied policy evaluations have the same basic difficulty: there are many possible comparison units but only a few time periods. San Francisco’s 2018 Sugary Drinks Distributor Tax is a useful example. The annual panel used here contains only thirteen periods, with seven pre-treatment years and six post-treatment years. With so little pre-treatment history, even a reasonably large donor pool does not make it easy to choose among competing causal panel estimators. The empirical design treats San Francisco County as the single treated unit because both the adult obesity outcome and the socioeconomic covariates are observed at the county or county-region level. The donor pool uses the eligible reported California county or county-region units in the final balanced analysis panel, and Alameda County is excluded because Berkeley and Oakland, both located within Alameda County, also adopted sugar-sweetened beverage taxes while the corresponding city-level indices are unavailable. The application is therefore a single-treated-county design.

A growing literature develops causal learning methods for policy evaluation with panel data. Fixed effects, random effects, difference-in-differences, synthetic control, matrix completion, and related estimators may each be reasonable tools. Which one works best depends on features of the untreated outcome process that the researcher does not observe directly: linearity, factor structure, and donor representability. With a long panel, repeated validation over time can help sort out which one is best. However, in a short panel, the validation set may contain only a few pre-treatment periods, making cross-validation unreliable in two ways.

First, the limited number of validation observations makes validation errors noisy, so small differences in validation performance may reflect sampling variation rather than genuine differences in counterfactual accuracy. Second, cross-validation in this setting differs from random-split cross-validation in ordinary regression. Counterfactual prediction must respect the time ordering of the panel: later pre-treatment outcomes cannot be used to tune predictions for earlier periods. The validation set is therefore typically a terminal block of pre-treatment periods. Because there is usually only one useful chronological split,

the researcher cannot average validation performance over many random folds, which limits the usual variance-reduction benefit of cross-validation.

The short time dimension also limits standard ensemble tools. Ordinary stacking estimates continuous weights on the candidate predictions by minimizing validation loss. With eight candidate causal estimators and only a few validation periods, unrestricted stacking asks a sparse time-series holdout block to identify many weights. The resulting optimization can be weakly identified or ill-conditioned. Thus the problem is not only unstable model selection. Conventional ensemble weighting can also overfit the small validation window when the method library is large relative to the number of validation periods, as in the San Francisco soda-tax application.

To address these two challenges—unstable model selection and the infeasibility of traditional ensemble learning methods—this paper proposes *Perturbed Cross-Validation* (PCV) for short-panel policy evaluation. PCV repeatedly perturbs the observed panel, evaluates each candidate estimator on the short chronological validation window, records the estimator with the lowest validation error, and converts these winning frequencies into ensemble weights. The final counterfactual prediction is a weighted average of the predictions from all candidate estimators. Theory and simulations show that this ensemble approach can improve counterfactual accuracy relative to CV-based selection when validation rankings are noisy.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 presents the short-panel setup, the PCV algorithm, perturbation design, and candidate estimators. Section 4 states the theoretical results, with proofs collected in the Appendix. Section 5 reports simulation evidence. Section 6 presents the soda-tax application. Section 7 concludes.

2 Literature Review and Positioning

The paper draws on four related literatures: causal panel methods, causal machine learning, stacking and model averaging, and perturbation-based stabilization. Each supplies part of

the problem PCV addresses; none resolves the short-panel selection problem on its own.

2.1 Causal panel methods and counterfactual prediction

The potential-outcomes framework defines the causal estimand by comparing observed treated outcomes with unobserved untreated potential outcomes (Rubin, 1974; Imbens and Rubin, 2015). In panel policy settings, difference-in-differences and two-way fixed effects remain common because they are transparent and easy to implement, although their interpretation can be sensitive to heterogeneous treatment effects, treatment timing, and violations of parallel trends (Card and Krueger, 1994; Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021). Synthetic control constructs a weighted donor counterfactual for a treated unit and has become a central tool for comparative case studies (Abadie and Gardeazabal, 2003; Abadie, Diamond, and Hainmueller, 2010; Abadie, 2021). Generalized synthetic control and interactive fixed-effects methods add latent-factor structure (Bai, 2009; Xu, 2017), while synthetic difference-in-differences combines weighting with DiD-style comparisons (Arkhangelsky et al., 2021). Matrix completion methods treat untreated potential outcomes as missing entries in a panel matrix and estimate them under low-rank regularization (Athey, Bayati, Doudchenko, Imbens, and Khosravi, 2021; Agarwal, Dahleh, Shah, and Shen, 2021). These estimators expand the set of credible counterfactuals, but they sharpen rather than settle the question of which one to use in a given application.

The empirical question we study has a direct antecedent. Zhao et al. (2025) apply the augmented synthetic control method within a target-trial framework to estimate the effect of the same one-cent-per-ounce California beverage taxes on adult obesity prevalence, using city-level estimates from the California Health Interview Survey. Their estimates are imprecise and statistically indistinguishable from zero, with a San Francisco point estimate near zero and a short pre-treatment window that they identify as the source of the imprecision. That study uses a single counterfactual estimator from the synthetic-control family; the contribution here is methodological—a disciplined way to choose among and combine many such estimators when the pre-treatment window is short—and the application serves to illustrate the method and to benchmark its conclusion against this independent estimate.

2.2 Causal machine learning and validation

Causal machine learning stresses that flexible prediction tools must be adapted to causal targets rather than used mechanically (Mullainathan and Spiess, 2017; Athey and Imbens, 2019). Double/debiased machine learning, for example, combines flexible nuisance estimation with orthogonal scores to reduce regularization bias (Chernozhukov et al., 2018). The panel problem studied here differs in emphasis: the object is not a high-dimensional nuisance function but the untreated counterfactual trajectory for a small number of treated units. Validation must therefore be time-based and conducted within the pre-treatment window. When that window is short, validation losses are noisy and method rankings are unstable.

A direct benchmark is Zhang and Yang (2015), who study cross-validation for selecting a model-selection procedure and show that it can consistently select the best procedure once the validation sample is large enough relative to the training sample. That result clarifies when winner-take-all CV is justified. The present setting is the complementary one: the number of validation periods is fixed by the short pre-treatment dimension, and adding donor units does not create additional pre-treatment years. The large-validation condition therefore fails, and the relevant goal becomes stabilization rather than consistent selection of a single winner.

2.3 Stacking, Super Learner, and model averaging

Stacked generalization and stacked regressions combine predictions from multiple models using validation-based weights (Wolpert, 1992; Breiman, 1996b). Super Learner provides an oracle-motivated framework for cross-validation-based ensemble learning (van der Laan, Polley, and Hubbard, 2007), and related work studies model aggregation and forecast combination (Yang, 2001, 2004; Yao et al., 2018). In causal panel settings, Athey, Bayati, Imbens, and Qu (2019) propose an ensemble approach that combines regression, synthetic-control, and matrix-completion predictions using stacked-regression weights. Their results show that combining counterfactual estimators can improve prediction relative to relying

on a single method.

The short-panel setting studied here creates a different problem. Conventional stacking requires enough validation information to estimate continuous combination weights. In short-panel causal applications, the binding validation dimension is the number of pre-treatment periods, not the number of donor units. When the validation block is short relative to the method library, unrestricted stacking is weakly identified: the validation criterion carries too little information to discipline a full weight vector. PCV keeps the ensemble idea but replaces direct estimation of continuous stacking weights with perturbation-based winning frequencies.

2.4 Instability and perturbation-based stabilization

PCV is closest in spirit to work on unstable model selection, perturbation aggregation, and noise-injection regularization. Bagging improves unstable predictors by aggregating predictions across bootstrap perturbations (Breiman, 1996a), and Breiman’s analysis of instability in model selection shows that procedures whose outputs shift sharply under small data perturbations can often be improved by aggregation (Breiman, 1996c). Noise injection on the scale of unexplained variation acts as a regularizer (Bishop, 1995). In related work, Zhang and Shen (2015) use data perturbation for adaptive procedure selection and recommend perturbation scales comparable to residual variation in regression settings. Cross-validation is a standard tool for prediction assessment and model choice (Stone, 1974; Shao, 1993), but its finite-sample behavior depends heavily on the validation design, and surveys document the associated bias-variance tradeoffs and variability of selection rules (Arlot and Celisse, 2010). Stability-based approaches further show that prediction-tuned selection rules can be unstable and that stability is informative in its own right (Lim and Yu, 2016). PCV brings this idea to causal panel method choice by perturbing the training outcomes, repeating the same chronological validation comparison, and converting repeated validation rankings into ensemble weights.

3 Causal Machine Learning Methodology: Setting, PCV, and Perturbation Scale

We observe a panel $\{Y_{it}\}$ for units $i \in \{1, \dots, N\}$ and periods $t \in \{1, \dots, T\}$. Unit $i = 1$ is the treated unit. Treatment begins after a pre-treatment window of length T_0 :

$$W_{it} = \mathbf{1}\{i = 1, t > T_0\}.$$

Potential outcomes are $Y_{it}(0)$ and $Y_{it}(1)$, and the observed outcome is

$$Y_{it} = (1 - W_{it})Y_{it}(0) + W_{it}Y_{it}(1). \quad (1)$$

The target is the treated unit's untreated post-treatment counterfactual trajectory, $\{Y_{1t}(0) : t > T_0\}$, and the corresponding effects

$$\tau_t = Y_{1t}(1) - Y_{1t}(0), \quad t = T_0 + 1, \dots, T.$$

A library of K candidate causal panel estimators produces predictions

$$\hat{Y}_{1t}^{(k)}(0), \quad k = 1, \dots, K, \quad t > T_0.$$

The difficulty is not fitting any one estimator. It is deciding how to choose or combine estimators when the pre-treatment history is short. The formal setup keeps the number of pre- and post-treatment periods fixed and small, while allowing the donor pool to be large. In the soda-tax application, $T_0 = 7$, $T_1 = 6$, and $T = 13$. Even if three pre-treatment periods are set aside for validation, the validation sample remains much smaller than the candidate library used below. Reserving more periods would leave too little training history for several estimators to be feasible or stable.

Let $\mathcal{T} \subset \{1, \dots, T_0\}$ denote the training block and let $\mathcal{V} \subset \{1, \dots, T_0\}$ denote the validation block. The split is chronological: $t < t'$ for all $t \in \mathcal{T}$ and $t' \in \mathcal{V}$. In practice, $\mathcal{V}$ is the

terminal block of the pre-treatment period. A policy-evaluation exercise cannot use later pre-treatment outcomes to tune predictions for earlier periods in the way a random-split regression exercise might. There is typically only one useful validation split, since expanding-window or rolling-window cross-validation is often infeasible when the pre-treatment period is short. Let $m = |\mathcal{V}|$ denote the length of this terminal validation block.

Ordinary CV selection chooses a single estimator,

$$\hat{k}_{CV} = \arg \min_{k \leq K} CV_k,$$

where CV_k is the validation loss of method k . Standard stacking instead chooses continuous weights $w \in \Delta_K$ by solving

$$\hat{w}_{stack} \in \arg \min_{w \in \Delta_K} \sum_{t \in \mathcal{V}} \left(Y_{1t} - \sum_{k=1}^K w_k \hat{Y}_{1t}^{(k)}(0) \right)^2, \quad \Delta_K = \{w \geq 0 : \sum_{k=1}^K w_k = 1\}.$$

Stacking is attractive when the validation sample is informative and when many splits can average out split-specific noise. Here, however, m is small, the candidate library can be large, and the chronological structure leaves only one usable validation block. Unrestricted stacking then estimates a high-dimensional weight vector from a sparse set of time-series validation observations. PCV is designed to avoid that ill-conditioned weight-estimation problem while still using the information in the holdout window.

3.1 Perturbed Cross-Validation

PCV does not create new time splits, and it never uses future outcomes to validate past predictions. In each iteration, the observed panel is perturbed by multiplying outcomes by mean-one random factors. Each candidate estimator is then fitted on the perturbed pre-treatment training block and evaluated on the same terminal validation block. The method that wins is recorded, and the winning frequencies across perturbations become the ensemble weights.

Let $b = 1, \dots, B$ index perturbation draws and let σ denote the outcome-noise scale used

for the perturbation. For a perturbation multiplier $a \geq 0$, define the mean-one multiplicative factor

$$R_{it}^{(b)}(a) = \frac{1 + (a\sigma\xi_{it}^{(b)})^2}{1 + a^2\sigma^2}, \quad \xi_{it}^{(b)} \stackrel{iid}{\sim} \mathcal{N}(0, 1), \quad (2)$$

and construct perturbed outcomes as

$$Y_{it}^{\star(b)} = Y_{it}R_{it}^{(b)}(a). \quad (3)$$

Because $E\{(\xi_{it}^{(b)})^2\} = 1$, the perturbation factor satisfies $E\{R_{it}^{(b)}(a)\} = 1$. The perturbation therefore preserves the observed outcome level on average while adding controlled multiplicative variation. The multiplier a is a temperature parameter. If a is too small, PCV is close to winner-take-all CV selection. If a is too large, the perturbation can swamp the validation signal and make the weights too diffuse.

3.2 Perturbation scale

The implementation uses the mean-preserving multiplicative perturbation in (2)–(3). In the simulations, σ is known by construction and is varied directly. In the soda-tax application, the same symbol σ denotes the scale used by the algorithm; it is set equal to the residual standard error from an auxiliary fixed-effect outcome regression.

Let Y_{it} denote the adult obesity rate and let X_{it} collect the county-level covariates used in the application. After excluding Alameda County and statewide aggregate observations, estimate the auxiliary two-way fixed-effect regression

$$Y_{it} = \alpha_i + \lambda_t + X_{it}'\gamma + u_{it} \quad (4)$$

on the pre-treatment county-year panel available for the analysis. In the empirical application, the perturbation scale is set to

$$\sigma = \left(\frac{1}{n-p} \sum_{i,t} \hat{u}_{it}^2 \right)^{1/2}, \quad (5)$$

where n is the number of county-year observations and p is the number of estimated regression coefficients, including the fixed effects. The auxiliary regression is used only to calibrate the perturbation scale. It is not one of the competing causal estimators and does not determine the final counterfactual.

The simulations report both $a = 1$ and $a = 0.5$, and we recommend $a = 0.5$ as the default scale for applied work (Section 5.4). These choices control the dispersion of the mean-one multiplicative perturbation through the product $a\sigma$. The scale is invariant to the units of the outcome and is tied to residual variation in the outcome process. A closely related perturbation scale is recommended by [Zhang and Shen \(2015\)](#) in the regression setting. The default is not claimed to be universally optimal; it is a fixed, conservative choice motivated by the instability problem in short pre-treatment panels and supported by the simulation evidence. The point is scale matching: the validation problem should be perturbed enough to reveal selection instability, but not so much that the validation signal is ignored.

3.3 Validation loss and winning frequencies

For draw b , define the validation loss of method k by

$$CV_k^{(b)}(a) = \frac{1}{|\mathcal{V}|} \sum_{t \in \mathcal{V}} \ell(Y_{1t}^{\star(b)}, \hat{Y}_{1t}^{\star(k,b)}),$$

where $\ell(a, b) = |a - b|$ in the implementation. Let

$$k_b^{\star}(a) = \arg \min_{k \leq K} CV_k^{(b)}(a).$$

The PCV weight for method k is the empirical winning frequency

$$\hat{\pi}_k(a) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{k_b^{\star}(a) = k\}. \quad (6)$$

The PCV counterfactual is

$$\hat{Y}_{1t}^{PCV}(0; a) = \sum_{k=1}^K \hat{\pi}_k(a) \hat{Y}_{1t}^{(k)}(0), \quad t > T_0. \quad (7)$$

The perturbation size affects the final estimator only through these stability weights. After the validation exercise, the candidate predictions $\hat{Y}_{1t}^{(k)}(0)$ are fitted on the full pre-treatment window and averaged using the PCV weights.

3.4 PCV algorithm

Algorithm 1 Perturbed Cross-Validation with Mean-Preserving Multiplicative Perturbations

Require: Panel data; candidate estimators $\{\omega_k\}_{k=1}^K$; pre-treatment window $\{1, \dots, T_0\}$; validation rule; loss ℓ ; perturbation scale σ ; perturbation multiplier $a \geq 0$; number of perturbation draws B .

- 1: Fit each ω_k on the unperturbed full pre-treatment window and store $\hat{Y}_{1t}^{(k)}(0)$ for $t > T_0$.
 - 2: Initialize $n_k \leftarrow 0$ for $k = 1, \dots, K$.
 - 3: **for** $b = 1, \dots, B$ **do**
 - 4: Draw $\xi_{it}^{(b)} \stackrel{iid}{\sim} \mathcal{N}(0, 1)$ and construct $R_{it}^{(b)}(a)$ and $Y_{it}^{\star(b)} = Y_{it} R_{it}^{(b)}(a)$ as in (2)–(3).
 - 5: **for** $k = 1, \dots, K$ **do**
 - 6: Fit method k on the perturbed pre-treatment training data.
 - 7: Compute $CV_k^{(b)}(a)$ on the perturbed validation window.
 - 8: **end for**
 - 9: Let $k_b^*(a) = \arg \min_k CV_k^{(b)}(a)$ and set $n_{k_b^*} \leftarrow n_{k_b^*} + 1$.
 - 10: **end for**
 - 11: Set $\hat{\pi}_k(a) = n_k/B$.
 - 12: Return $\hat{Y}_{1t}^{PCV}(0; a) = \sum_{k=1}^K \hat{\pi}_k(a) \hat{Y}_{1t}^{(k)}(0)$.
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Remark 1. *PCV weights are empirical frequencies of being the best performer under the mean-preserving multiplicative perturbation distribution calibrated by $a\sigma$. They are not posterior probabilities, and no likelihood is required. The perturbation scale σ is known in the*

simulations and estimated from auxiliary fixed-effect residual variation in the application.

3.5 Candidate estimators in the implementation

The candidate library contains eight panel estimators spanning classical and more recent approaches: fixed effects, random effects, synthetic control, matrix completion, horizontal elastic net, vertical elastic net, difference-in-differences, and an ADH-style synthetic-control estimator. The implementation reports two benchmarks: (i) **CV selection**, which chooses the method with the smallest pre-treatment holdout loss; and (ii) **PCV averaging**, which computes winning-frequency weights under mean-preserving multiplicative perturbations and averages the full-sample post-treatment predictions using those weights.

4 Theory: PCV Ensemble Learning for Short-Panel Causal Prediction

This section gives a compact theoretical justification for PCV in the short-panel regime. PCV is not an oracle, and the paper does not claim that it uniformly dominates cross-validation (CV) selection across all data-generating processes. The relevant case is more modest and more common in applications: the pre-treatment validation window is short, the validation ranking is noisy, and several candidate estimators contain useful but imperfect information.

The theoretical results are stated for the perturbation scheme introduced in Section 3. For each perturbation draw b , the observed outcome is multiplied by the mean-one factor

$$R_{it}^{(b)}(a) = \frac{1 + (a\sigma\xi_{it}^{(b)})^2}{1 + a^2\sigma^2}, \quad Y_{it}^{\star(b)} = Y_{it}R_{it}^{(b)}(a), \quad \xi_{it}^{(b)} \stackrel{iid}{\sim} \mathcal{N}(0, 1).$$

Candidate methods are re-estimated on the perturbed pre-treatment training block and evaluated on the same terminal validation block. The PCV weights are the empirical winning frequencies across perturbation draws. The final post-treatment counterfactual is then formed as the weighted average of the unperturbed candidate predictions fitted on the full

pre-treatment window.

The asymptotic sequence is a short-panel sequence:

$$N_0 \rightarrow \infty, \quad N_1, T_0, T_1, K \text{ fixed.}$$

Here N_0 is the donor-pool size, N_1 is the number of treated units, T_0 is the pre-treatment length, T_1 is the post-treatment length, and K is the number of candidate estimators. The number of validation periods, $m = |\mathcal{V}|$, is fixed and small. It cannot grow without shortening the training block, which must remain long enough for the candidate estimators to be estimable. Because the split must respect chronology, validation uses a single terminal pre-treatment block rather than many random folds. This is what separates the present setting from large-validation cross-validation theory. For example, [Zhang and Yang \(2015\)](#) show that CV can consistently select the best model-selection procedure once the validation sample is large enough relative to the training sample. Here the validation window cannot grow with N_0 , and the researcher cannot average over many random folds. Validation noise therefore need not vanish. The results below formalize this distinction and show why perturbation-smoothed combination can help in this short-pre-treatment-period regime.

4.1 Fixed validation and the limitation of winner-take-all selection

Let R_k denote the target post-treatment prediction risk of candidate method k and let CV_k denote its validation criterion. The following representation is a local abstraction of a fixed validation window:

$$CV_k = R_k + s\eta_k, \quad k = 1, \dots, K,$$

where $s \geq 0$ measures the scale of validation noise and η_k is a centered noise component. This representation is not intended as a full parametric model for the panel. It is a local way to capture the fact that, with only a few validation periods, the validation criterion is a noisy estimate of the target risk.

Theorem 1 (Fixed-validation limitation of CV selection). *Suppose $K < \infty$ and $CV_k =$*

$R_k + s\eta_k$. Let k_0 be the unique best method, and define $\Delta_k = R_k - R_{k_0} > 0$ for $k \neq k_0$.

(a) If each difference $\eta_k - \eta_{k_0}$ is sub-Gaussian with scale parameter ν^2 , then

$$\Pr\{\widehat{k}_{CV} \neq k_0\} \leq \sum_{k \neq k_0} \exp\left\{-\frac{\Delta_k^2}{2s^2\nu^2}\right\},$$

where $\widehat{k}_{CV} = \arg \min_k CV_k$. Hence CV selection is reliable when validation noise is small relative to the risk gaps.

(b) If $s > 0$ is fixed and for some $j \neq k_0$ the random variable $\eta_{k_0} - \eta_j$ has positive probability above Δ_j/s , then

$$\Pr(\widehat{k}_{CV} = j) > 0.$$

Consequently, when the validation noise does not vanish, winner-take-all CV selection need not consistently select the best method. In the two-method case, the expected excess risk of CV selection is

$$\mathbb{E}\{R_{\widehat{k}_{CV}} - R_{k_0}\} = \Delta_2 \Pr(\widehat{k}_{CV} = 2).$$

Theorem 1 is consistent with the result in [Zhang and Yang \(2015\)](#). If the validation sample grows and the validation noise scale s shrinks to zero, CV selection can recover the best procedure. In the short-panel sequence used here, m is fixed, so there is no comparable guarantee that s vanishes. The simulations follow this logic: when outcome noise is low, CV selection is already accurate; when outcome noise is moderate or high, the cost of winner-take-all selection becomes visible.

4.2 PCV as perturbation-smoothed selection

For each perturbation draw b , define the perturbed validation criterion exactly as in Section 3:

$$CV_k^{\star(b)}(a) = \frac{1}{m} \sum_{t \in \mathcal{V}} \ell\left(Y_{1t}^{\star(b)}, \widehat{Y}_{1t}^{\star(k,b)}\right),$$

where $Y_{it}^{\star(b)} = Y_{it}R_{it}^{(b)}(a)$ and $\hat{Y}_{1t}^{\star(k,b)}$ is the validation prediction produced by method k after fitting that method on the perturbed training block. Let

$$k_b^{\star}(a) = \arg \min_{1 \leq j \leq K} CV_j^{\star(b)}(a)$$

and define the conditional perturbation-winning probability

$$\pi_k(a) = \Pr\{k_b^{\star}(a) = k \mid \mathcal{D}\},$$

where $\mathcal{D}$ denotes the observed panel and the probability is over the multiplicative perturbation draws. The PCV weight is

$$\hat{\pi}_k(a) = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{k_b^{\star}(a) = k\}.$$

Theorem 2 (PCV weights as smoothed selection probabilities). *Fix the observed data $\mathcal{D}$ and a perturbation multiplier $a \geq 0$. Suppose the B multiplicative perturbation draws are conditionally independent and identically distributed, and ties in the perturbed validation criterion occur with conditional probability zero. Then for each k ,*

$$\mathbb{E}\{\hat{\pi}_k(a) \mid \mathcal{D}\} = \pi_k(a), \quad \text{Var}\{\hat{\pi}_k(a) \mid \mathcal{D}\} = \frac{\pi_k(a)\{1 - \pi_k(a)\}}{B},$$

and

$$\hat{\pi}_k(a) \xrightarrow{p} \pi_k(a) \quad \text{as } B \rightarrow \infty.$$

If the unperturbed chronological validation criterion has a unique minimizer $\hat{k}_{CV}$ and the candidate validation losses are continuous in the perturbation scale at $a = 0$, then $\pi_{\hat{k}_{CV}}(a) \rightarrow 1$ as $a \downarrow 0$.

The theorem gives the operational interpretation of PCV. The weights are not unrestricted stacking weights and are not posterior model probabilities. They are stability weights: the fraction of times a candidate method wins when the same chronological valida-

tion problem is perturbed by mean-preserving multiplicative factors. Small perturbations reproduce ordinary CV selection, while very large perturbations can wash out the validation signal and produce overly diffuse weights. The useful region lies between these extremes: large enough to reflect selection uncertainty, but not so large that the validation signal is overwhelmed.

4.3 When combination can improve on selection

The paper does not claim that PCV universally dominates selection. Such a claim would require unrealistic restrictions. The argument is conditional: combination can improve on selection when candidate estimators have complementary errors, and PCV inherits this advantage when its perturbation-smoothed weights approximate a risk-improving convex combination.

Let e_k denote the post-treatment prediction error of method k , stacked over the treated unit-periods being evaluated. For $w \in \Delta_K = \{w \in \mathbb{R}_+^K : \sum_k w_k = 1\}$, define the ensemble error and risk by

$$e(w) = \sum_{k=1}^K w_k e_k, \quad R(w) = \mathbb{E}\{\|e(w)\|_2^2\}.$$

Let ι_k denote the k th unit vector in $\mathbb{R}^K$, so $R(\iota_k)$ is the risk of the k th individual estimator.

Theorem 3 (Combination advantage and PCV risk). *Assume $\mathbb{E}\|e_k\|_2^2 < \infty$ for all k .*

- (a) *Suppose two candidate estimators have equal risk $v > 0$ and covariance $\mathbb{E}(e'_1 e_2) = \rho v$ with $\rho < 1$. Then the equal-weight combination $w = (1/2, 1/2)$ has risk*

$$R(w) = \frac{v}{2}(1 + \rho) < v,$$

so a convex combination strictly dominates both individual estimators.

- (b) *More generally, suppose there exists $w^* \in \Delta_K$ and $\delta > 0$ such that*

$$R(w^*) \leq \min_{1 \leq k \leq K} R(\iota_k) - \delta.$$

If the PCV weights satisfy $\hat{w} \xrightarrow{p} w^*$ and the risk function is continuous at w^* , then for every $\epsilon \in (0, \delta)$,

$$\Pr \left\{ R(\hat{w}) \leq \min_k R(\iota_k) - \delta + \epsilon \right\} \rightarrow 1.$$

In particular, PCV asymptotically inherits the advantage of the risk-improving convex combination it approximates.

The first part states the basic reason averaging can outperform selection: imperfectly correlated prediction errors can offset one another. The second gives the corresponding PCV statement, which is conditional rather than universal. If perturbation-smoothed validation places weight near a useful combination, PCV can improve on individual selection. If one method is clearly dominant and validation noise is small, Theorem 1 implies that both CV selection and PCV concentrate on that method, so the gain from combination is limited. That is the pattern found in the low-noise simulations.

4.4 The limiting PCV weights and the optimal perturbation scale

Theorem 3(b) is conditional: it assumes the PCV weights converge to a risk-improving combination w^* but does not characterize the weights themselves. This subsection closes that gap. Working in the local validation model of Theorem 1, we characterize the limiting PCV weights as a function of the perturbation scale and, in the two-method case, derive the risk-optimal scale in closed form.

Conditional on the observed panel D , method k has realized validation criterion $c_k = CV_k = R_k + s\eta_k$. PCV perturbs this criterion: for replication b ,

$$CV_k^{\star(b)}(a) = c_k + a G_k^{(b)},$$

where the perturbation-induced shift $G^{(b)} = (G_1^{(b)}, \dots, G_K^{(b)})$ is drawn i.i.d. across b from a distribution with continuous density and exchangeable coordinates, and the multiplier a

plays the role of the perturbation scale. The conditional winning probability is

$$\pi_k(a) = \Pr\{c_k + aG_k \leq c_j + aG_j \ \forall j \mid D\},$$

and the PCV ensemble risk along the resulting weight path is $R(\pi(a)) = \pi(a)^\top A \pi(a)$ with $A_{jk} = \mathbb{E}(e_j^\top e_k)$ the post-treatment error Gram matrix.

Theorem 4 (Limiting weights and optimal scale). *Assume $\mathbb{E}\|e_k\|_2^2 < \infty$ for all k and that the perturbation law is exchangeable with continuous density.*

- (a) (Smoothed selection.) *The weight path interpolates between selection and uniform weighting: $\pi_k(a) \rightarrow \mathbf{1}\{c_k = \min_j c_j\}$ as $a \downarrow 0$ and $\pi_k(a) \rightarrow 1/K$ as $a \rightarrow \infty$, and $a \mapsto \pi(a)$ is continuous on $(0, \infty)$.*
- (b) (Two-method optimal scale.) *Let $K = 2$, $\Delta = c_2 - c_1$, and suppose $\Delta > 0$ (method 1 is the CV winner; $\Delta < 0$ is symmetric). Let F be the c.d.f. of $G_1 - G_2$, so $F(0) = \frac{1}{2}$. Then $\pi_1(a) = F(\Delta/a)$, which decreases strictly from 1 to $\frac{1}{2}$ as a increases. The risk $R(\pi)$ is a convex quadratic with unconstrained minimizer*

$$\pi^* = \frac{A_{22} - A_{12}}{A_{11} - 2A_{12} + A_{22}}, \quad A_{11} - 2A_{12} + A_{22} = \mathbb{E}\|e_1 - e_2\|_2^2.$$

If $\pi^ \in (\frac{1}{2}, 1)$, the PCV risk $R(\pi_1(a))$ has the unique interior minimizer*

$$a^* = \frac{\Delta}{F^{-1}(\pi^*)} \in (0, \infty), \quad R(\pi_1(a^*)) = R(\pi^*) < R_1 = R(\pi_1(0)).$$

If $\pi^ \leq \frac{1}{2}$ the risk is decreasing in a (optimal $a \rightarrow \infty$); if $\pi^* \geq 1$ it is increasing in a (optimal $a \rightarrow 0$).*

Three implications follow. First, part (b) closes the gap in Theorem 3: it gives explicit conditions (π^* interior) under which the PCV weights move strictly toward the risk-minimizing combination, so the antecedent $\hat{w} \rightarrow w^*$ is delivered rather than assumed. Second, the optimal scale $a^* = \Delta/F^{-1}(\pi^*)$ is proportional to the realized validation gap $\Delta = (R_2 - R_1) + s(\eta_2 - \eta_1)$, tying the scale to the risk gap and the error structure and

explaining why an intermediate scale is optimal; because a^* depends on unknown population quantities, however, it is used here as a theoretical rationale for the fixed default rather than as a tuning target (Section 5.4). Third, the value of perturbing separates into an *averaging* gain $R_1 - R(\pi^*)$, present whenever $\pi^* < 1$ (complementary errors), and a *selection-stability* gain that grows with the validation-noise scale s , because larger s makes the CV winner more often wrong. As $s \rightarrow 0$ the selection-stability gain vanishes and PCV reduces to selection up to the averaging term, matching the low-noise simulations.

Remark 2 (General K). *For $K > 2$ the winning probabilities and the risk $R(\pi(a))$ admit no closed form, since the optimal scale depends on the full geometry of A and of the perturbation law. The qualitative prediction of part (b)—an interior risk-minimizing scale in the noisy regime—is nonetheless visible in the simulations: in Figure 1 the average gain of PCV over CV selection rises and then falls as the perturbation scale increases in the linear and nonlinear designs, peaking at an intermediate scale. A general closed-form optimal scale is left open.*

4.5 Additional implications

Proposition 1 (Ill-conditioning of unrestricted stacking in short panels). *Let m be the number of validation observations and K the number of candidate methods. The unrestricted stacking problem*

$$\min_{w \in \Delta_K} \|y - Xw\|_2^2$$

is not strictly convex on the affine hull of the simplex whenever $m < K - 1$. Therefore, with fewer validation observations than free weights, the validation loss cannot identify a unique interior stacking weight vector.

This proposition explains why unrestricted stacking is not the main benchmark. In short-panel designs with a limited validation block and a relatively large method library, the validation sample cannot reliably support direct estimation of a full vector of continuous stacking weights. PCV uses the validation sample more cautiously, constructing stability weights through repeated perturbed selections.

Remark 3 (No universal dominance). *No non-oracle ensemble rule can uniformly dominate the best individual estimator over all possible data-generating processes. If one candidate estimator is exactly correct and the ensemble assigns positive weight to another estimator with nonzero error, the ensemble has positive risk while the correct individual estimator has zero risk. The claim here is therefore regime-specific. PCV should be close to selection when validation is easy, and it can improve performance when validation is noisy and candidate errors are complementary.*

Remark 4 (Perturbation scale in empirical work). *The empirical analysis uses the same mean-preserving multiplicative perturbation as the simulations, with the outcome-noise scale σ set equal to the residual scale from the auxiliary fixed-effect outcome regression. Specifically,*

$$R_{it}(a) = \frac{1 + (a\sigma\xi_{it})^2}{1 + a^2\sigma^2}, \quad Y_{it}^* = Y_{it}R_{it}(a), \quad \xi_{it} \sim \mathcal{N}(0, 1).$$

The auxiliary regression used to estimate σ is only a calibration device; it is not a competing causal estimator. The perturbation is positive and preserves the outcome level in expectation because $E R_{it}(a) = 1$. The values $a = 1$ and $a = 0.5$ reported in the simulations are not treated as universally optimal; we recommend $a = 0.5$ as a fixed default rather than a tuned quantity, for the reasons set out in [Section 5.4](#).

5 Monte Carlo Simulations

This section evaluates PCV in short panels and compares its performance with ordinary cross-validation. Stacked regression is not included as a benchmark because, with only two validation periods and eight candidate estimators, unrestricted stacking weights are weakly identified. The simulations therefore focus on whether perturbation-based ensemble weights can stabilize validation-based estimator choice when the validation block is too small to support reliable estimation of continuous stacking weights.

5.1 Simulation design

Each simulated panel contains $N = 40$ units observed for $T = 10$ periods. For each unit-period observation, we observe one outcome variable, Y_{it} , and $K = 5$ covariates, $X_{1it}, \dots, X_{Kit}$. Unit 1 is treated in the final five periods, leaving five pre-treatment periods for training and validation. The terminal pre-treatment validation block contains two periods, and the remaining three earlier pre-treatment periods form the training block. We focus on a two-period validation block for two practical reasons. A one-period validation block performs worse than the two-period block across the simulation settings, while a three-period validation block leaves too little training history and makes some candidate estimators inestimable in some replications.

The simulations consider three data-generating processes (DGPs). Let $i = 1, \dots, N$ index units, $t = 1, \dots, T$ index time periods, and $k = 1, \dots, K$ index covariates. Unit $i = 1$ is the treated unit. The covariates are generated in the same way across all three DGPs. For each unit i and covariate k , a latent covariate process Z_{kit} follows the ARMA process

$$\begin{aligned} Z_{kit} &= 0.8897Z_{ki,t-1} - 0.4858Z_{ki,t-2} + u_{kit} - 0.2279u_{ki,t-1} + 0.2488u_{ki,t-2}, \\ u_{kit} &\sim N(0, 0.1796). \end{aligned}$$

The observed covariate is

$$X_{kit} = Z_{kit} + v_{it}, \quad v_{it} \sim N(0, 1),$$

where v_{it} is common across the five covariates within a unit-period observation. The three DGPs differ in the untreated mean $\mu_{it}(0)$. Throughout, let $\eta_i \sim N(0, 1)$.

Linear DGP. For all units including the treated unit, the untreated mean is

$$\mu_{it}(0) = 0.5X_{2it} + 1.5X_{4it} + \eta_i.$$

Nonlinear DGP. For all units, the untreated mean is

$$\mu_{it}(0) = 0.25X_{2it}^2 + 1.5X_{4it} + \eta_i.$$

Synthetic-control DGP. For the donor units, the untreated mean is first generated using the linear DGP:

$$\mu_{it}(0) = 0.5X_{2it} + 1.5X_{4it} + \eta_i, \quad i \neq 1.$$

The treated unit's untreated mean is then generated by

$$\mu_{1t}(0) = 0.5\mu_{3t}(0) + 0.5\mu_{5t}(0).$$

In all three DGPs, treatment is assigned only to unit 1 in the final five post-treatment periods:

$$D_{it} = \mathbf{1}\{i = 1, t \geq 6\}.$$

The heterogeneous treatment effect is

$$\tau_{it} = D_{it} \left(\sin(X_{1it}^2) + 0.5X_{3it}^2 \right).$$

The observed outcome is therefore

$$Y_{it} = \mu_{it}(0) + \tau_{it} + \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, \sigma^2).$$

For each design, the noise standard deviation is set to $\sigma \in \{0.5, 1, 2\}$. We run 100 simulation replications for each combination of DGP and noise level, always using a two-period terminal pre-treatment validation block.

The candidate library contains eight counterfactual estimators: fixed effects (FE), random effects (RE), synthetic control (SC), nuclear-norm matrix completion (MC), horizontal elastic net (EN), vertical elastic net (EN-T), difference-in-differences (DID), and the

Abadie–Diamond–Hainmueller synthetic-control (ADH) estimator. The horizontal elastic-net estimator predicts missing outcomes using time-series information across periods. The vertical elastic-net estimator applies the same regularized-regression logic after transposing the outcome matrix and therefore uses cross-sectional donor information.

5.2 Comparison rules and performance measures

The benchmark is ordinary cross-validation selection. For each validation block, all candidate estimators are trained on the pre-validation history and evaluated on the terminal pre-treatment validation periods. CV selection chooses the estimator with the lowest validation error and uses that estimator to construct the post-treatment counterfactual prediction.

For each perturbation replication, the candidate estimators are re-estimated on the perturbed training block and evaluated on the perturbed terminal validation block. PCV records the estimator with the smallest validation error in each replication and converts the resulting winning frequencies into ensemble weights. The final PCV counterfactual is the weighted average of the candidate post-treatment predictions, with each estimator’s weight equal to its winning frequency across 100 perturbations. We report results for perturbation multipliers $a = 0.5$ and $a = 1$.

Performance is measured by the mean absolute error of the estimated untreated counterfactual path for the treated unit in the post-treatment periods. For each simulation replication, we compute the average absolute error over the five post-treatment periods. Table 1 reports the average MAE for CV selection and for PCV with perturbation multipliers $a = 0.5$ and $a = 1$. The table also reports the gain relative to CV selection,

$$\text{Gain} = \text{MAE}_{CV} - \text{MAE}_{PCV},$$

so positive values indicate that PCV improves on CV selection. The reported p -values come from paired two-sided t -tests across the 100 simulation replications.

Table 1: Simulation evidence with short chronological validation blocks

DGP	σ	CV-select	PCV (0.5σ)	PCV (σ)	Gain (0.5σ)	p (0.5σ)	Gain (σ)	p (σ)
Linear	0.5	0.712	0.680	0.695	0.032	0.304	0.017	0.677
Linear	1.0	1.263	1.151	1.187	0.112	0.015	0.077	0.176
Linear	2.0	2.078	1.906	1.977	0.172	0.001	0.101	0.085
Nonlinear	0.5	0.840	0.828	0.850	0.011	0.695	-0.011	0.760
Nonlinear	1.0	1.300	1.197	1.227	0.103	0.004	0.073	0.122
Nonlinear	2.0	2.055	1.884	1.893	0.171	0.009	0.162	0.017
Synthetic control	0.5	1.738	1.659	1.568	0.078	0.023	0.170	< 0.001
Synthetic control	1.0	1.926	1.826	1.753	0.100	0.055	0.173	0.006
Synthetic control	2.0	2.684	2.403	2.325	0.281	< 0.001	0.359	< 0.001

Notes: The table reports mean absolute error for the treated unit’s untreated counterfactual path over the five post-treatment periods. All rows use a two-period terminal pre-treatment validation block. Gain is defined as $MAE_{CV} - MAE_{PCV}$, so positive values favor PCV. The reported p-values are from paired two-sided t -tests across simulation replications. The candidate library contains eight estimators: fixed effects (FE), random effects (RE), synthetic control (SC), nuclear-norm matrix completion (MC), horizontal elastic net (EN), vertical elastic net (EN-T), difference-in-differences (DID), and the Abadie–Diamond–Hainmueller synthetic-control (ADH) estimator.

5.3 Simulation results

The simulations show that PCV is most useful in the settings for which it is designed: short validation blocks and noisy validation rankings. With the validation block fixed at two periods, ordinary CV selection is generally more stable when the noise level is low. As the noise level increases, validation rankings become less reliable, and the value of smoothing those rankings through PCV becomes more apparent.

Across the three DGPs, PCV with $a = 0.5$ delivers positive average gains in all reported cells. PCV with $a = 1$ also improves on CV selection in nearly all cells, although the larger perturbation is less stable in the low-noise nonlinear design. This pattern supports the interpretation of PCV as a stabilization device rather than a universally dominant replacement for CV. The method reduces the risk of committing to a single estimator when validation evidence is noisy, while remaining close to ordinary CV when the validation ranking is informative.

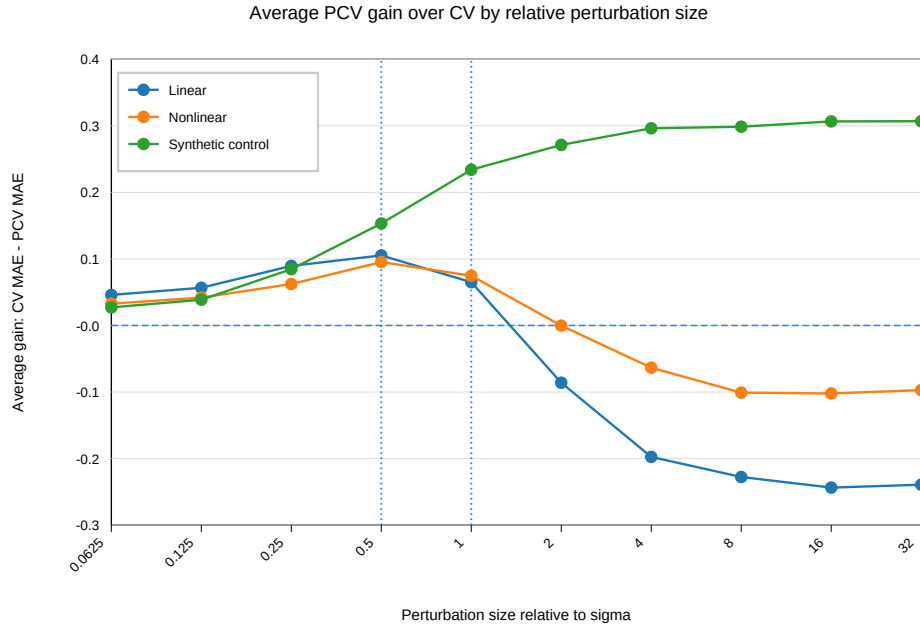


Figure 1: Average PCV gain over CV selection by perturbation scale with validation size fixed at two periods. The vertical axis reports $MAE_{CV} - MAE_{PCV}$, averaged across simulation replications and noise levels. Positive values favor PCV. The dotted vertical lines mark $a = 0.5$ and $a = 1$.

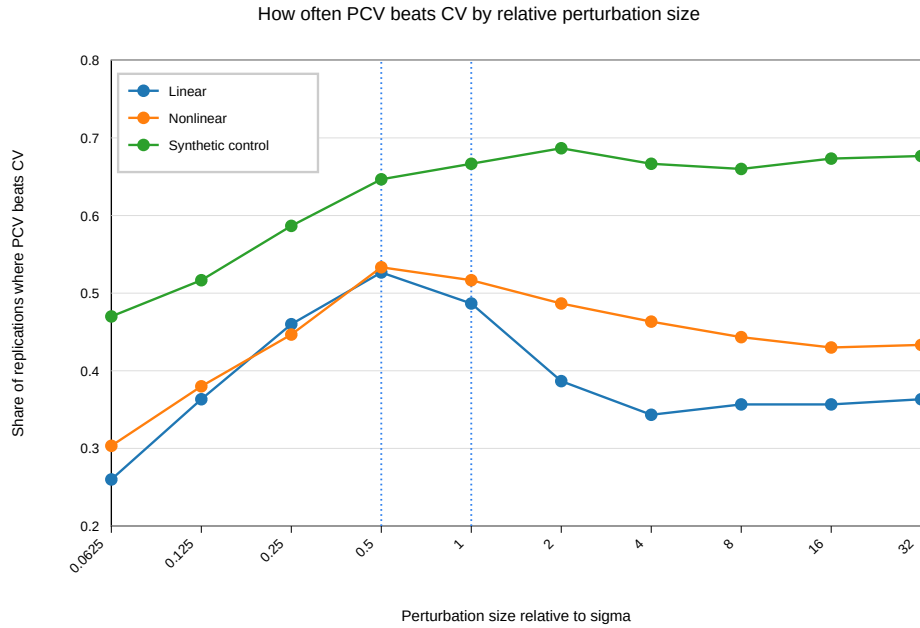


Figure 2: Share of simulation replications in which PCV outperforms CV selection with validation size fixed at two periods. The dotted vertical lines mark $a = 0.5$ and $a = 1$.

Figure 1 summarizes the sensitivity of PCV to the perturbation scale. Very small perturbations behave similarly to ordinary CV selection. For the linear and nonlinear DGPs, gains are largest near the recommended scales and then decline when the perturbations become large. This non-monotone pattern is the empirical counterpart of Theorem 4: the gain peaks at an intermediate scale because the ensemble risk $R(\pi(a))$ has an interior minimum, with very small a reverting to CV selection and very large a over-diffusing the weights toward uniform averaging. In the synthetic-control DGP, larger perturbations continue to stabilize selection over the scale range considered here. Figure 2 shows the share of replications in which PCV outperforms CV selection. The win-rate evidence reinforces the same point: PCV’s gains are not driven only by a few extreme replications, but by more frequent improvements when validation rankings are noisy.

Overall, the simulations support two conclusions. First, single-estimator selection by ordinary CV can be unstable in short panels, and PCV improves on CV selection across the simulation designs considered here. Second, a single fixed perturbation scale is sufficient: $a = 0.5$ delivers positive gains in every reported cell and across all three designs, and the gain is not sharply sensitive to the exact value of a in the useful range (Figure 1). We therefore recommend $a = 0.5$ as the default perturbation scale for applied work, with $a = 1$ reported throughout as a robustness comparison.

5.4 Why the perturbation scale is fixed rather than tuned

A natural question is whether a should be selected from the data rather than fixed. We deliberately treat a as a fixed default rather than a tuning parameter, for four reasons.

First, the short pre-treatment window that motivates PCV also makes data-driven selection of a unreliable. Choosing a empirically would require comparing perturbation scales on held-out validation data, but in a short panel that validation block is exactly the scarce resource the method is built around: it cannot be enlarged without starving the training block. Worse, selecting a by validation performance recurses the very problem PCV is designed to solve. Ordinary cross-validation is an unreliable way to choose among estimators in short panels; choosing a by the same short validation block is unreliable for the same

reason. Tuning a would therefore reintroduce the short-sample selection noise that PCV exists to smooth out.

Second, the simulations indicate that tuning is unnecessary. A single fixed scale, $a = 0.5$, produces positive gains across all three designs and all noise levels, and Figure 1 shows that performance is flat over a range of intermediate scales rather than sharply peaked. The cost of using a fixed default instead of the best scale for a particular design is therefore small, while the risk of a poorly estimated data-driven scale is not.

Third, this is consistent with the theory rather than in tension with it. Theorem 4 characterizes a risk-optimal scale a^* , but $a^* = \Delta/F^{-1}(\pi^*)$ depends on the population risk gap and the error structure—quantities that cannot be estimated precisely in a short panel. The theorem is best read as an explanation of *why* an intermediate scale works, not as a recipe for estimating one. Attempting to plug in noisy short-sample estimates of a^* would forfeit the robustness that the fixed default provides.

Fourth, fixing a keeps PCV usable and protects the inference. Applied causal-panel work benefits from methods with few tuning knobs: each additional researcher degree of freedom is one more choice that must be justified and that a reader may suspect was made with the result in view. In the single-treated-unit design this is not merely a convenience. If a were chosen by its effect on the treated unit’s post-treatment gap, the placebo p -value would no longer be valid, because the procedure applied to the treated unit would differ from the procedure applied to the placebo units. Fixing a in advance, and applying the identical scale to every unit, is what makes the placebo comparison legitimate. For these reasons we fix $a = 0.5$ and report robustness across a as a diagnostic, not as a tuning exercise.

6 Application: San Francisco’s Sugar-Sweetened Beverage Tax

This section applies PCV to the effect of San Francisco County’s sugar-sweetened beverage tax on adult obesity prevalence. The application illustrates the empirical problem that motivates the method: several counterfactual estimators are plausible, but the pre-treatment

history is short, so validation-based selection is potentially unstable and conventional stacking is weakly disciplined. The headline finding is a null: the PCV counterfactual implies no detectable effect of the tax on adult obesity prevalence, a conclusion that is robust to the perturbation scale and consistent with independent synthetic-control estimates for the same policy (Zhao et al., 2025).

6.1 Institutional setting and data

San Francisco voters approved a one-cent-per-ounce excise tax on sugar-sweetened beverages in November 2016, and the tax took effect on January 1, 2018. The empirical analysis treats San Francisco County as the treated unit and uses annual county- and county-region-level observations on adult obesity prevalence and socioeconomic covariates. Because San Francisco is coterminous with San Francisco County, the policy jurisdiction aligns with the reported outcome used in the analysis.

The outcome is adult obesity prevalence, modeled on the logit scale: with p_{it} the reported prevalence, the modeling outcome is $Y_{it} = \log\{p_{it}/(1 - p_{it})\}$, and all predictions are mapped back to the prevalence scale for reporting. Modeling a bounded proportion on the logit scale is standard and avoids predictions that ignore the $[0, 1]$ boundary; it is adopted as a fixed default for all units and is not selected on the basis of the treated-unit outcome.¹ The obesity-prevalence series and the socioeconomic covariates are drawn from *The California Health Interview Survey (CHIS)* (<https://healthpolicy.ucla.edu/our-work/california-health-interview-survey-chis>), for the years 2011–2023. Because CHIS is a survey, obesity status is based on respondents’ self-reported height and weight, which is a further source of measurement error in addition to sampling variation. The combining of sparsely populated counties into multi-county regions follows the source’s own reporting. A complete variable dictionary and the raw and cleaned data are provided in the replication repository (see the Data and Code Availability statement). Because the

¹In an earlier version of this analysis the outcome was the combined overweight-or-obesity rate, modeled on the raw (untransformed) scale. Because that rate sits near the upper boundary of the unit interval, the untransformed counterfactual fit the treated unit poorly and produced a large but spurious estimated gap. Modeling the bounded proportion on the logit scale and using the obesity outcome—the outcome examined in the directly comparable literature—resolves this; see the discussion in Section 7.

small-county prevalence estimates are survey-based, they carry substantial sampling noise; this is material for interpreting the width of the placebo distribution below.

The donor pool consists of eligible reported California county or county-region units in the final balanced analysis panel that were not directly exposed to the San Francisco tax during the study period. Some sparsely populated counties are combined into multi-county regions in the original source, so the analysis contains 43 reported geographic units, including San Francisco (so the placebo comparison uses 42 untreated donor units). Alameda County is excluded because Berkeley and Oakland, both within Alameda County, adopted beverage taxes whose effects predate 2018 (Berkeley’s tax took effect in 2015), creating contamination across the whole panel rather than only the post-treatment window. The candidate counterfactual estimators are trained on the pre-treatment period and used to predict San Francisco’s untreated obesity path after the tax.

6.2 Candidate estimators and PCV implementation

The candidate library contains eight counterfactual estimators: fixed effects, random effects, synthetic control, nuclear-norm matrix completion, horizontal elastic net, vertical elastic net, difference-in-differences, and the Abadie–Diamond–Hainmueller synthetic-control estimator. The horizontal elastic-net estimator uses time-series information across periods, while the vertical elastic-net estimator applies the same regularized-regression procedure to the transposed outcome matrix and therefore uses cross-sectional donor information. We refer to these estimators as EN and EN-T, respectively.

PCV uses the same chronological validation logic as in the simulations. With seven pre-treatment periods (2011–2017), weights are formed by training the candidate estimators on the earliest pre-treatment block (2011–2014) and evaluating them on the terminal pre-treatment validation block (2015–2017); the winning frequencies under mean-preserving perturbations of these outcomes become the ensemble weights. The final counterfactual is then formed by refitting the candidates on the full pre-treatment window (2011–2017) and averaging their post-treatment (2018–2023) predictions with those weights. The post-treatment outcomes play no role in weight selection, so the reported gap is genuinely held

out from the selection step. The scale parameter σ is set equal to the residual standard error from an auxiliary pre-treatment fixed-effect regression on the logit-scale outcome. For perturbation replication b , the perturbed outcome is

$$Y_{it}^{\star(b)} = Y_{it} R_{it}^{(b)}(a), \quad R_{it}^{(b)}(a) = \frac{1 + (a\sigma\xi_{it}^{(b)})^2}{1 + a^2\sigma^2}, \quad \xi_{it}^{(b)} \sim N(0, 1),$$

applied with the identical scale to the treated unit and to every placebo unit. The primary specification uses the recommended default $a = 0.5$, and we report the placebo distribution across a as a robustness check rather than as a tuning exercise (Section 5.4). For each replication, PCV records the estimator with the smallest validation error and converts the resulting winning frequencies into ensemble weights.

6.3 PCV weights

In the corrected obesity analysis the weights concentrate on a single dominant estimator rather than spreading across the library, so the combination prediction coincides with the CV-selection prediction: for San Francisco the selected estimator and the ensemble produce identical counterfactual paths in every post-treatment year. A single estimator—fixed effects—wins essentially every perturbed validation contest, so its PCV weight is one. This places San Francisco in the stable-ranking regime of Section 4, in which PCV coincides with winner-take-all CV selection. It is corroborated by the perturbation-robustness reported below: San Francisco’s estimated gap is invariant to the perturbation multiplier a , exactly the behavior expected when one estimator dominates the validation contest at every scale.

This concentration is specific to San Francisco, not a general feature of the procedure. Table 2 reports, for each of the 43 units, how many candidate estimators receive positive PCV weight at $a = 0.5$. For 26 of the 43 units (60%), PCV forms a strict combination of two or more estimators rather than selecting one; for the remaining 17, including San Francisco, a single estimator dominates the perturbed validation contest and PCV coincides with CV selection. The best estimator is moreover not the same across units: all eight candidates are the top-weighted choice for at least one unit, so there is no fixed estimator that selection

could safely default to. PCV is therefore adaptive—it combines when the validation ranking is unstable and reduces to selection when one estimator is decisively best, as it is for San Francisco. The efficiency gain from combining is quantified where the truth is known, in the simulations of Section 5; on a single treated unit the counterfactual is unobserved, so the application illustrates the procedure rather than measuring its gain.

Table 2: Concentration of PCV weights across the 43 geographic units ($a = 0.5$)

Estimators with positive weight	Units	Share
1 (PCV = CV selection)	17	40%
2	17	40%
3	7	16%
4	2	5%
Any combination (≥ 2)	26	60%

Notes: Number of candidate estimators receiving positive PCV weight for each unit, at perturbation scale $a = 0.5$. San Francisco is one of the 17 units in the first row, where a single estimator (fixed effects) wins every perturbed validation contest, so PCV equals CV selection. For the other 26 units PCV forms a strict combination. Across the 43 units, all eight candidate estimators are the top-weighted choice for at least one unit.

6.4 Estimated counterfactual path

Figure 3 plots San Francisco’s observed adult obesity rate and the PCV no-tax counterfactual for the post-treatment years 2018–2023. The counterfactual tracks the observed series closely, and the post-treatment gaps are small. They are also not monotone in the direction of a tax effect: in the first three post-treatment years (2018–2020) the observed rate sits slightly *above* the counterfactual, and the small positive gaps appear only in 2021–2023, a window that coincides with the COVID-19 pandemic and the associated disruption to survey-based prevalence measurement. A tax effect would be expected to appear at implementation and persist; this pattern does not.

For reporting, define the post-treatment counterfactual gap in year t as

$$\hat{G}_t = \hat{Y}_{SF,t}(0) - Y_{SF,t},$$

where $Y_{SF,t}$ is the observed adult obesity rate in San Francisco and $\hat{Y}_{SF,t}(0)$ is the PCV

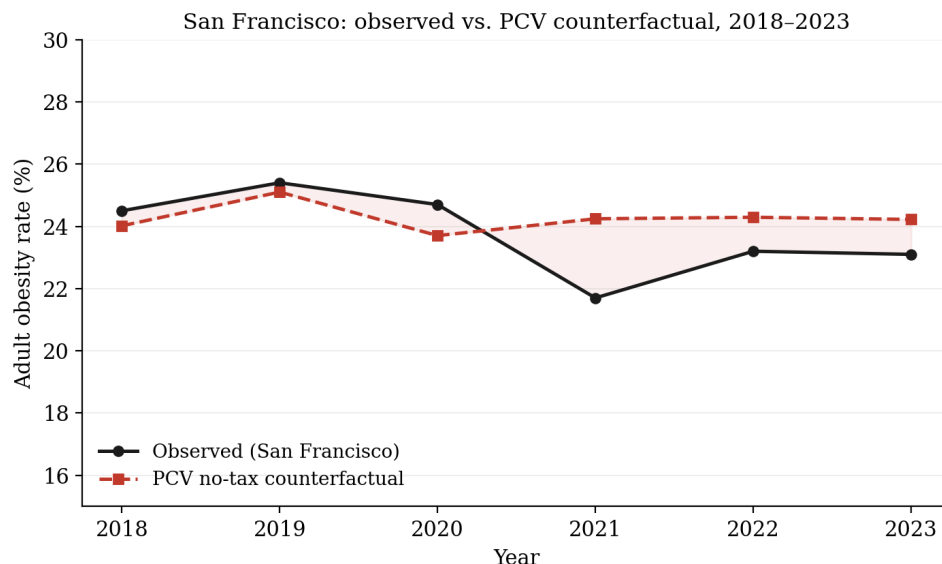


Figure 3: Observed and PCV no-tax counterfactual adult obesity rates for San Francisco County, 2018–2023. The gap is defined as the counterfactual rate minus the observed rate. The figure shows the post-treatment period; the pre-treatment fit of the counterfactual is discussed in the text.

counterfactual (on the prevalence scale). Positive values indicate that the observed rate is below the predicted no-tax counterfactual. The year-by-year gaps are -0.5 , -0.3 , and -1.0 percentage points in 2018–2020 and $+2.5$, $+1.1$, and $+1.1$ percentage points in 2021–2023, averaging 0.50 percentage points over the post-treatment period. The average is small relative to the year-to-year variation in the underlying series, is of the wrong sign in the immediate post-treatment years, and turns positive only in the pandemic window. The next subsection benchmarks this magnitude against the gaps produced for untreated units.

6.5 Placebo evidence

To assess whether the San Francisco gap is unusual relative to gaps that arise for untreated units, we conduct a leave-one-unit-out placebo exercise within the same donor pool. Each untreated county or county-region unit is reassigned placebo treatment beginning on January 1, 2018, and the full PCV procedure is re-estimated using the remaining eligible units (San Francisco, the genuinely treated unit, is excluded from every placebo donor pool).

This produces a placebo distribution of post-treatment gaps for units not actually treated by the San Francisco tax.

San Francisco’s average gap lies near the center of this distribution rather than in its tail; in fact it sits almost exactly at the placebo median (0.50 versus a median of 0.50 and a mean of 0.9, with the distribution ranging from about -9.5 to $+10.4$ percentage points). In the directional comparison, San Francisco ranks 22nd of 43 units, giving a randomization-style p -value of 0.51; in the absolute-value comparison it is among the units with the *smallest* gaps, ranking 40th of 43 ($p = 0.93$). Neither comparison is close to statistical significance. The conclusion is moreover invariant to the perturbation multiplier: San Francisco’s gap of 0.50 percentage points is unchanged across $a \in \{0.125, 0.25, 0.5, 1\}$, consistent with the single-estimator concentration of its PCV weights, and at the larger scale $a = 2$ —well above the recommended default of $a = 0.5$ —the gap rises only to 1.9 percentage points and remains statistically indistinguishable from the placebo distribution (directional $p = 0.44$). Figure 4 shows the full distribution of average gaps with San Francisco highlighted.

Two features of the placebo distribution deserve comment. First, it is wide: average gaps range from about -9.5 to $+10.4$ percentage points, and single-year gaps are larger still. This dispersion is driven mainly by measurement noise in small-county prevalence estimates rather than by the method, and it is the reason San Francisco’s small gap is unremarkable. Second, because the units differ sharply in how precisely their counterfactuals can be fit, the raw-gap comparison gives equal footing to well-measured and poorly-measured units; a natural refinement is to normalize each unit by its own pre-treatment fit, for example through the ratio of post-treatment to pre-treatment fit error (the post/pre RMSPE ratio) or by population-weighting the placebo distribution. Because San Francisco already lies at the center of the raw-gap distribution and among the best-fit units, these refinements would only reinforce the null.

6.6 Donor-pool definition

The empirical analysis uses the eligible reported California county and county-region units in the final balanced analysis panel, after excluding Alameda County because of potential

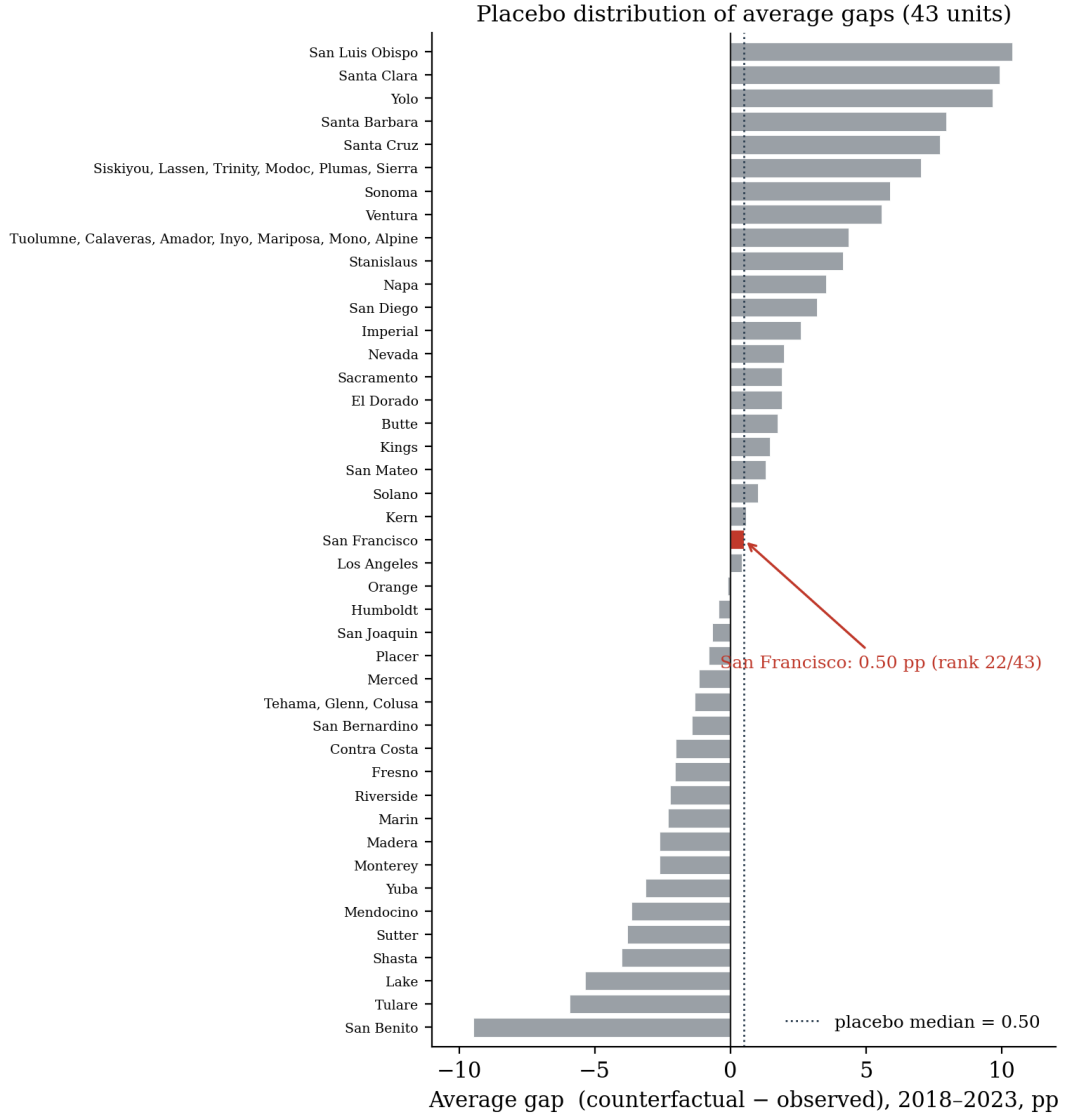


Figure 4: Placebo distribution of average post-treatment gaps for adult obesity prevalence, 2018–2023. Bars report each reported county or county-region unit’s average value of $\hat{Y}_{it}(0) - Y_{it}$, with San Francisco highlighted. San Francisco lies near the center of the distribution.

contamination from Berkeley and Oakland. The unit of analysis is therefore the geographic unit as reported in the source publication. For most of the state, this unit is an individual county. For some sparsely populated areas, the source combines multiple counties into one reported region. The placebo p-values should therefore be read as comparisons against all eligible reported geographic units in the data, not as comparisons against 58 separate California counties.

6.7 Discussion

The San Francisco application illustrates both the usefulness and the limits of PCV. The method provides a disciplined way to construct a counterfactual path when several panel estimators are plausible but the pre-treatment validation window is short. Applied to adult obesity prevalence on the logit scale, PCV produces a counterfactual that fits San Francisco well and implies a post-treatment gap that is small, of the wrong sign in the immediate post-tax years, and statistically indistinguishable from the gaps produced for untreated units. We therefore find no detectable effect of the tax on adult obesity prevalence.

This null is consistent with the independent evidence. [Zhao et al. \(2025\)](#), using an augmented synthetic control on city-level obesity prevalence, report a San Francisco estimate near zero with a wide confidence interval, and the broader literature finds robust declines in sugar-sweetened-beverage purchases and consumption but small or undetectable effects on body-weight outcomes. That PCV—applied carefully, with selection held out from the reported gap and the perturbation scale fixed in advance—recovers a conclusion in line with these estimates is itself informative: in a setting where naive specification choices produced a large spurious effect, the disciplined procedure did not manufacture a false positive.

The contrast with an earlier specification is instructive for the method. Modeling the combined overweight-or-obesity rate on the raw scale—a proportion near the upper boundary of the unit interval—yielded a counterfactual that fit the treated unit poorly and a placebo distribution in which San Francisco appeared extreme, producing an apparent double-digit “effect.” That apparent effect was an artifact of poor counterfactual fit, not a treatment signal: it dissolved once the bounded proportion was modeled on the logit

scale and the outcome examined in the comparable literature was used. This sensitivity to specification is exactly the instability that motivates the paper, and it underscores that the placebo benchmark and the pre-treatment fit of the counterfactual must be examined before any gap is read as causal. The finite placebo distribution, the noisiness of small-county prevalence estimates, and the post-2018 overlap with the COVID-19 pandemic remain important qualifications.

7 Discussion and Conclusion

This paper develops perturbed cross-validation for short-panel policy evaluation. The motivation is simple. When the pre-treatment history is short, ordinary cross-validation has too little validation information to choose reliably among plausible counterfactual estimators, while unrestricted stacking asks too much of the same small validation block. PCV sits between the two: it preserves the chronological validation structure that causal panel applications require, but replaces winner-take-all selection with an ensemble learner.

The simulations and the San Francisco application both show that this distinction matters. PCV helps most when validation rankings are noisy, and it stays close to ordinary cross-validation when one estimator is consistently favored. In the application, the San Francisco case falls in the second regime—one estimator dominates the validation contest—so PCV reduces to selection there, and the resulting counterfactual implies no detectable effect of the tax on adult obesity prevalence, a conclusion robust to the perturbation scale and consistent with independent synthetic-control estimates for the same policy.

A second limitation concerns external validation of the counterfactual. With only seven pre-treatment periods, a held-out pre-period cross-validation diagnostic cannot reliably certify a six-year-ahead extrapolation: there is no way to hold out a validation horizon comparable to the deployment horizon within so short a window. This is a genuine constraint, but it is distinct from—and should not be confused with—the internal validation step inside PCV. The unreliability of single-split cross-validation in short panels is the premise that motivates PCV; the method’s response is to perturb and aggregate rather than to trust

one noisy split. Inference in the application therefore rests on the cross-unit placebo distribution, which exploits the donor dimension where the data are informative and does not require additional pre-treatment periods.

The deeper open problem is inference after data-driven method choice. Standard errors and placebo distributions usually condition on a chosen estimator, yet in practice that estimator is itself selected or tuned on the same short pre-treatment history, and ignoring this step can understate uncertainty. Data perturbation offers a natural way forward: the perturbations used here to stabilize estimator choice could also trace the variability of the selected counterfactual path, the stability of the ensemble weights, and the uncertainty contributed by the selection step itself. Developing valid perturbation-based inference for short-panel causal designs is, in our view, the most useful next step.

Data and Code Availability

All code, raw data, and cleaned analysis-ready data are publicly available at <https://github.com/eliozhangsu/Adaptive-Ensemble-Learning-for-Short-Panel-Causal-Policy-Evaluation>. The repository contains the data-cleaning, simulation, and application scripts, the generated tables and figures, and a variable dictionary documenting the source and definition of each variable; the main tables and figures can be reproduced by running `master.R`.

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Appendix

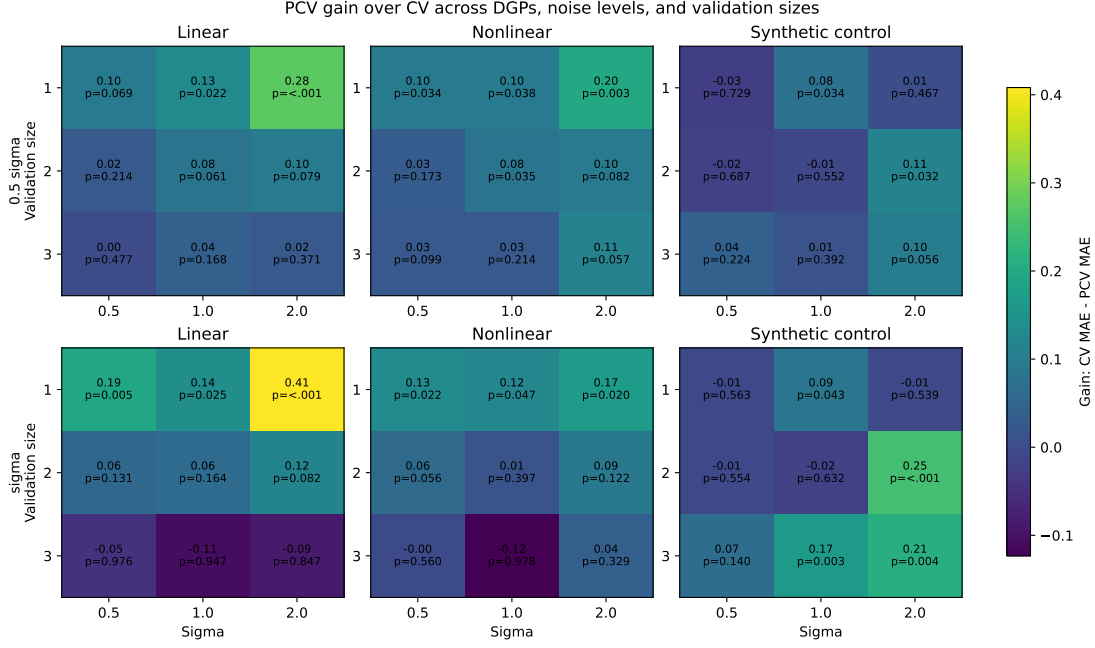


Figure 5: Simulation gains and paired significance tests for the recommended perturbation scales

Notes: Each cell reports the gain $MAE_{CV} - MAE_{PCV}$ and the paired t -test p -value for the corresponding DGP, noise level, and validation size. Positive gains indicate that PCV has lower average MAE than winner-take-all CV selection. The upper row uses perturbation size 0.5σ ; the lower row uses perturbation size σ .

Proof of Theorem 1

We prove each part separately.

For part (a), let k_0 be the unique minimizer of R_k . CV selects a method different from k_0 only if at least one inferior method has a validation criterion no larger than that of k_0 :

$$\{\hat{k}_{CV} \neq k_0\} \subseteq \bigcup_{k \neq k_0} \{CV_k \leq CV_{k_0}\}.$$

Using $CV_k = R_k + s\eta_k$, the event $CV_k \leq CV_{k_0}$ is equivalent to

$$R_k + s\eta_k \leq R_{k_0} + s\eta_{k_0}.$$

Since $\Delta_k = R_k - R_{k_0} > 0$, this can be rewritten as

$$s(\eta_k - \eta_{k_0}) \leq -\Delta_k,$$

or, equivalently,

$$\eta_{k_0} - \eta_k \geq \Delta_k/s.$$

When $s = 0$, the probability of this event is zero because $R_k > R_{k_0}$. For $s > 0$, the assumed sub-Gaussian tail bound gives

$$\Pr(CV_k \leq CV_{k_0}) = \Pr(\eta_{k_0} - \eta_k \geq \Delta_k/s) \leq \exp \left\{ -\frac{\Delta_k^2}{2s^2\nu^2} \right\}.$$

Taking a union bound over all $k \neq k_0$ yields

$$\Pr\{\widehat{k}_{CV} \neq k_0\} \leq \sum_{k \neq k_0} \Pr(CV_k \leq CV_{k_0}) \leq \sum_{k \neq k_0} \exp \left\{ -\frac{\Delta_k^2}{2s^2\nu^2} \right\}.$$

This proves part (a).

For part (b), fix $j \neq k_0$. CV selects j whenever $CV_j \leq CV_{k_0}$ and, if there are more than two methods, CV_j is also no larger than the remaining criteria. The theorem states a sufficient non-vanishing error condition for the pairwise comparison against the best method. In the two-method case this condition is exact. We have

$$CV_j \leq CV_{k_0} \iff R_j + s\eta_j \leq R_{k_0} + s\eta_{k_0} \iff \eta_{k_0} - \eta_j \geq \Delta_j/s.$$

By assumption,

$$\Pr(\eta_{k_0} - \eta_j \geq \Delta_j/s) > 0.$$

Thus the inferior method has positive probability of beating the best method in validation whenever s is fixed and positive. In the two-method case, this event is exactly $\{\widehat{k}_{CV} = j\}$, so $\Pr(\widehat{k}_{CV} = j) > 0$.

In the two-method case, the selected risk is

$$R_{\widehat{k}_{CV}} = R_{k_0} \mathbf{1}\{\widehat{k}_{CV} = k_0\} + R_j \mathbf{1}\{\widehat{k}_{CV} = j\}.$$

Subtracting R_{k_0} gives

$$R_{\widehat{k}_{CV}} - R_{k_0} = (R_j - R_{k_0}) \mathbf{1}\{\widehat{k}_{CV} = j\} = \Delta_j \mathbf{1}\{\widehat{k}_{CV} = j\}.$$

Taking expectations gives

$$\mathbb{E}\{R_{\widehat{k}_{CV}} - R_{k_0}\} = \Delta_j \Pr(\widehat{k}_{CV} = j).$$

Renaming $j = 2$ proves the displayed formula. Since the probability is positive under the stated condition, the expected excess risk is positive. This proves the theorem.

Proof of Theorem 2

Fix the observed panel $\mathcal{D}$ and the perturbation multiplier a . For each perturbation draw b , the algorithm draws an array $\xi^{(b)} = \{\xi_{it}^{(b)}\}$ and constructs

$$R_{it}^{(b)}(a) = \frac{1 + (a\sigma\xi_{it}^{(b)})^2}{1 + a^2\sigma^2}, \quad Y_{it}^{\star(b)} = Y_{it} R_{it}^{(b)}(a).$$

Conditional on $\mathcal{D}$, the only randomness in the perturbed validation criterion

$$CV_k^{\star(b)}(a) = \frac{1}{m} \sum_{t \in \mathcal{V}} \ell\left(Y_{1t}^{\star(b)}, \widehat{Y}_{1t}^{\star(k,b)}\right)$$

comes from the perturbation draw. Because the perturbation draws are conditionally independent and identically distributed, the induced winning indicators

$$Z_{bk}(a) = \mathbf{1}\{k = \arg \min_j CV_j^{\star(b)}(a)\}$$

are also conditionally independent and identically distributed across b . The no-ties assumption implies that exactly one method wins each perturbed validation exercise. Therefore $Z_{bk}(a)$ is a Bernoulli random variable with conditional success probability

$$\pi_k(a) = \Pr\{k = \arg \min_j CV_j^*(a) \mid \mathcal{D}\},$$

where $CV_j^*(a)$ denotes a generic perturbed validation criterion generated from one multiplicative perturbation draw.

The PCV weight is the sample mean of these indicators:

$$\hat{\pi}_k(a) = \frac{1}{B} \sum_{b=1}^B Z_{bk}(a).$$

Linearity of conditional expectation gives

$$\mathbb{E}\{\hat{\pi}_k(a) \mid \mathcal{D}\} = \frac{1}{B} \sum_{b=1}^B \mathbb{E}\{Z_{bk}(a) \mid \mathcal{D}\} = \pi_k(a).$$

Conditional independence gives

$$\text{Var}\{\hat{\pi}_k(a) \mid \mathcal{D}\} = \frac{1}{B^2} \sum_{b=1}^B \text{Var}\{Z_{bk}(a) \mid \mathcal{D}\} = \frac{\pi_k(a)\{1 - \pi_k(a)\}}{B}.$$

Since this conditional variance converges to zero as $B \rightarrow \infty$, Chebyshev's inequality implies

$$\hat{\pi}_k(a) - \pi_k(a) \xrightarrow{p} 0.$$

This proves the unbiasedness, variance formula, and Monte Carlo convergence of the PCV weights.

It remains to prove the limiting statement for small perturbations. Let $CV_k(0)$ denote the unperturbed chronological validation criterion and suppose it has a unique minimizer $\hat{k}_{CV}$. Let

$$g = \min_{j \neq \hat{k}_{CV}} \{CV_j(0) - CV_{\hat{k}_{CV}}(0)\} > 0.$$

Because $R_{it}^{(b)}(a) \rightarrow 1$ as $a \downarrow 0$ for every fixed perturbation draw, the perturbed outcomes converge to the unperturbed outcomes. By the assumed continuity of candidate validation losses in the perturbation scale at $a = 0$, for each fixed perturbation draw outside a probability-zero exceptional set,

$$CV_k^*(a) \rightarrow CV_k(0) \quad \text{for every } k.$$

Hence, for sufficiently small a , the perturbed validation gap between every $j \neq \hat{k}_{CV}$ and $\hat{k}_{CV}$ remains positive. Therefore the perturbed winner equals $\hat{k}_{CV}$ with probability tending to one, and

$$\pi_{\hat{k}_{CV}}(a) \rightarrow 1 \quad \text{as } a \downarrow 0.$$

This proves the theorem.

Proof of Theorem 3

For part (a), consider two candidate errors e_1 and e_2 satisfying

$$\mathbb{E}\|e_1\|_2^2 = v, \quad \mathbb{E}\|e_2\|_2^2 = v, \quad \mathbb{E}(e_1' e_2) = \rho v,$$

with $v > 0$ and $\rho < 1$. The equal-weight ensemble has error

$$e(w) = \frac{1}{2}e_1 + \frac{1}{2}e_2.$$

Its risk is

$$R(w) = \mathbb{E} \left\| \frac{e_1 + e_2}{2} \right\|_2^2.$$

Expanding the squared norm gives

$$\left\| \frac{e_1 + e_2}{2} \right\|_2^2 = \frac{1}{4}\|e_1\|_2^2 + \frac{1}{4}\|e_2\|_2^2 + \frac{1}{2}e_1' e_2.$$

Taking expectations yields

$$R(w) = \frac{1}{4}v + \frac{1}{4}v + \frac{1}{2}\rho v = \frac{v}{2}(1 + \rho).$$

Because $\rho < 1$,

$$\frac{v}{2}(1 + \rho) < v.$$

Since both individual risks equal v , the equal-weight combination strictly dominates both individual estimators. This proves part (a).

For part (b), finite second moments imply that $R(w)$ is a finite quadratic function of w .

To see this, define the $K \times K$ matrix A by

$$A_{jk} = \mathbb{E}(e'_j e_k).$$

Then

$$R(w) = \mathbb{E} \left\| \sum_{k=1}^K w_k e_k \right\|_2^2 = \mathbb{E} \left\{ \sum_{j=1}^K \sum_{k=1}^K w_j w_k e'_j e_k \right\} = \sum_{j=1}^K \sum_{k=1}^K w_j w_k A_{jk} = w' A w.$$

Because K is finite and A has finite entries, $R(w) = w' A w$ is continuous on the simplex.

By assumption,

$$R(w^*) \leq \min_k R(\iota_k) - \delta.$$

Let $\epsilon \in (0, \delta)$. Since R is continuous at w^* , there exists a radius $r > 0$ such that

$$\|w - w^*\| < r \quad \Rightarrow \quad |R(w) - R(w^*)| < \epsilon.$$

Since $\hat{w} \xrightarrow{p} w^*$,

$$\Pr(\|\hat{w} - w^*\| < r) \rightarrow 1.$$

On this event,

$$R(\hat{w}) \leq R(w^*) + \epsilon \leq \min_k R(\iota_k) - \delta + \epsilon.$$

Therefore

$$\Pr \left\{ R(\hat{w}) \leq \min_k R(\iota_k) - \delta + \epsilon \right\} \rightarrow 1.$$

This proves the theorem.

Proof of Theorem 4

Part (a). Fix the data D , so $c_1, \dots, c_K$ are constants. For $a > 0$, $\arg \min_j \{c_j + aG_j\} = \arg \min_j \{c_j/a + G_j\}$ has the same distribution-free structure throughout. As $a \downarrow 0$, $aG_j \rightarrow 0$ for every fixed draw, so for any draw outside the probability-zero set where the minimizer of $\{c_j\}$ is tied, $\arg \min_j \{c_j + aG_j\} \rightarrow \arg \min_j c_j$; dominated convergence gives $\pi_k(a) \rightarrow \mathbf{1}\{c_k = \min_j c_j\}$. As $a \rightarrow \infty$, $c_j/a \rightarrow 0$, so $\arg \min_j \{c_j/a + G_j\} \rightarrow \arg \min_j G_j$ for almost every draw, and exchangeability of G with a continuous density gives $\Pr\{G_k = \min_j G_j\} = 1/K$, hence $\pi_k(a) \rightarrow 1/K$. Continuity of $a \mapsto \pi_k(a)$ on $(0, \infty)$ follows from continuity of the map $a \mapsto \Pr\{c_k + aG_k \leq c_j + aG_j \ \forall j\}$, which holds by dominated convergence because the boundary events $\{c_k + aG_k = c_j + aG_j\}$ have probability zero for each a under the continuous density.

Part (b). With $K = 2$,

$$\pi_1(a) = \Pr\{c_1 + aG_1 \leq c_2 + aG_2\} = \Pr\{a(G_1 - G_2) \leq c_2 - c_1\} = \Pr\{G_1 - G_2 \leq \Delta/a\} = F(\Delta/a).$$

Since F is strictly increasing (continuous positive density) and $\Delta > 0$, as a increases Δ/a decreases from $+\infty$ to 0, so $\pi_1(a)$ decreases strictly from 1 to $F(0) = \frac{1}{2}$, and $a \mapsto \pi_1(a)$ is a strictly decreasing bijection from $(0, \infty)$ onto $(\frac{1}{2}, 1)$.

Writing π for the weight on method 1,

$$R(\pi) = \pi^2 A_{11} + 2\pi(1-\pi)A_{12} + (1-\pi)^2 A_{22}, \quad R''(\pi) = 2(A_{11} - 2A_{12} + A_{22}) = 2\mathbb{E}\|e_1 - e_2\|_2^2 \geq 0,$$

so R is convex, and strictly convex unless $e_1 = e_2$ almost surely. Setting $R'(\pi) = 0$ gives the unconstrained minimizer $\pi^* = (A_{22} - A_{12})/(A_{11} - 2A_{12} + A_{22})$.

Consider $\Phi(a) := R(\pi_1(a))$. Since $\pi_1(\cdot)$ is a strictly decreasing bijection onto $(\frac{1}{2}, 1)$ and

R is strictly convex with vertex at π^* :

- If $\pi^* \in (\frac{1}{2}, 1)$, there is a unique a^* with $\pi_1(a^*) = \pi^*$, namely $a^* = \Delta/F^{-1}(\pi^*)$ (using $\pi_1(a) = F(\Delta/a)$). As a crosses a^* , $\pi_1(a)$ crosses π^* , so Φ decreases then increases; a^* is the unique minimizer, with $\Phi(a^*) = R(\pi^*)$. Because R is strictly convex and $\pi^* \neq 1$, $R(\pi^*) < R(1) = A_{11} = R_1 = \Phi(0^+)$.
- If $\pi^* \leq \frac{1}{2}$, then $\pi_1(a) \geq \frac{1}{2} \geq \pi^*$ for all a , and R is increasing on $[\pi^*, \infty)$; since π_1 decreases toward $\frac{1}{2}$, Φ is decreasing in a , with infimum approached as $a \rightarrow \infty$.
- If $\pi^* \geq 1$, then $\pi_1(a) \leq 1 \leq \pi^*$ and R is decreasing on $(-\infty, \pi^*]$; since π_1 decreases in a , Φ increases in a , minimized as $a \rightarrow 0$.

This proves part (b) and the theorem. $\square$

$$\mathcal{A} = \{w \in \mathbb{R}^K : \mathbf{1}'w = 1\}$$

be the affine hull of the simplex, and let

$$\mathcal{T} = \{v \in \mathbb{R}^K : \mathbf{1}'v = 0\}$$

be the tangent space of feasible directions in the affine hull. The dimension of $\mathcal{T}$ is $K - 1$.

The stacking objective is

$$Q(w) = \|y - Xw\|_2^2.$$

For any $w \in \mathcal{A}$ and any $v \in \mathcal{T}$, consider $w(a) = w + av$. Then

$$Q\{w(a)\} = \|y - Xw - aXv\|_2^2.$$

The second derivative along this path is

$$\frac{d^2}{da^2}Q\{w(a)\} = 2\|Xv\|_2^2.$$

Strict convexity on $\mathcal{A}$ requires $\|Xv\|_2^2 > 0$ for every nonzero $v \in \mathcal{T}$.

However, X is an $m \times K$ matrix, so $\text{rank}(X) \leq m$. If $m < K - 1 = \dim(\mathcal{T})$, the linear map $v \mapsto Xv$ from $\mathcal{T}$ into $\mathbb{R}^m$ cannot be injective. Therefore there exists a nonzero $v \in \mathcal{T}$ such that $Xv = 0$. Along any feasible segment $w + av \in \Delta_K$,

$$Q(w + av) = \|y - Xw - aXv\|_2^2 = \|y - Xw\|_2^2 = Q(w).$$

Thus the objective is flat in a nonzero feasible direction and cannot be strictly convex on the affine hull. Consequently, the validation loss cannot identify a unique interior stacking weight vector when $m < K - 1$. This proves the proposition.

Moments and Scale of the Multiplicative Perturbation

The implementation uses the mean-preserving multiplicative perturbation

$$R_{it}(a) = \frac{1 + (a\sigma\xi_{it})^2}{1 + a^2\sigma^2}, \quad Y_{it}^* = Y_{it}R_{it}(a), \quad \xi_{it} \sim \mathcal{N}(0, 1).$$

In the simulations, σ is known. In the empirical application, σ is set equal to the residual standard error from the auxiliary fixed-effect outcome regression used to calibrate the perturbation scale. Conditional on the data,

$$\mathbb{E}\{R_{it}(a)\} = 1, \quad \mathbb{E}(Y_{it}^* \mid \mathcal{D}) = Y_{it}.$$

Since ξ_{it}^2 has variance 2,

$$\text{Var}\{R_{it}(a)\} = \frac{2a^4\sigma^4}{(1 + a^2\sigma^2)^2}, \quad \text{Var}(Y_{it}^* - Y_{it} \mid \mathcal{D}) = Y_{it}^2 \frac{2a^4\sigma^4}{(1 + a^2\sigma^2)^2}.$$

Thus the perturbation is centered in multiplicative form: it preserves the observed outcome level in expectation while introducing positive, scale-calibrated variation in the training and validation outcomes. This section is not a proof of causal identification. It only records the moment properties of the perturbation device used to measure the stability of validation rankings.